Figures of Reasoning: Do Rhetorical Schemes Shape LLM Reasoning? Evidence from Prompting, Probing, and Activation Steering

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Abstract

Rhetorical figures are formal structures that carry communicative and argumentative meaning, yet are almost entirely absent from research on large language models (LLMs). Borrowing from AI safety interpretability research, which uses linear probes to detect whether models encode properties such as honesty or deception, this project asks three questions: **(a) can constraining an LLM to reason in a specific rhetorical form improve reasoning quality**, **(b) are those forms encoded in the model's internal representations**, and **(c) can those representations be causally steered?**

Using Llama-2-7b-chat on StrategyQA, a benchmark requiring multi-step binary reasoning, we prompt the model to reason using three figures: **enumeratio** (sequential numbered steps), **parison** (grammatically parallel clauses), and **chiasmus** (A-B-B-A structural inversion). We measure answer accuracy and faithfulness, defined as whether the reasoning chain logically entails the stated answer. We find that prompting with enumeratio and parison improves accuracy over baseline, rhetorical conditions are geometrically encoded in mid-to-late layer activations, and that steering activations along the parison direction improves both accuracy and faithfulness.

Motivation and Background

Rhetorical schemes are **form-function alignments** defined by their **material form** (phonological, orthographical, morpholexical, or syntactic), each aligned with a **communicative function**: Semantic-Feature Promotion, Reciprocal Specification, Reciprocal Energy, Irrelevance of Order/Rank, Subclassification, and Reject-Replace [1].

Because their **form**, schemes are computationally more **detectable** than tropes [1]. Because of their **functions**, they are directly relevant to argument and **reasoning** — Harris argues schemes "can get at meaning in a way that is beyond current text mining and NLU tools" [1], yet remain "almost universally ignored by ... computational linguists" [1].

This project explores both these callouts: if their functions provide valuable argumentative meaning, can prompting LLMs to use them improve reasoning? And if their form is detectable, can we find evidence of that encoding in the model's activation space?

Enumeratio: Explicitly numbered reasoning steps (1. 2. 3.) Each numbered step must be stated before the next, structuring the reasoning chain rather than leaving inferential jumps implicit.

Parison: Parallel clause structure with repeated grammatical form across reasoning steps. Fahnestock [2] argues parallel form "epitomizes the argument; it is the perfect expression of an induction based on examples". Repeating the same grammatical structure signals that each step belongs to the same category of evidence, grouping them into a unified argument.

Chiasmus: A–B ... B–A reasoning structure; may realise multiple chiastic subtypes. Harris [3] identifies the chiastic form as enacting Reciprocal Energy and Comprehensiveness. inversion stages two competing considerations and resolves their tension, mapping naturally onto binary reasoning tasks.

The probing and steering methodology is adapted from interpretability research. In 2023, Marks and Tegmark [4] demonstrated that truth-relevant properties are linearly encoded in LLM activation space and steerable. We apply the same framework to rhetorical structure.

Experimental Setup

Model: Llama-2-7b-chat-hf · **Dataset**: StrategyQA (200 questions, balanced yes/no) · **Judge**: Claude Haiku. *Each of the 200 questions was run under four prompt conditions:*

Figure	Prompt
Baseline	Answer the question by thinking **step by step.**
Enumeratio	Answer the question using **numbered reasoning steps.**
Parison	Answer the question using **parallel sentence structures** where possible, where each reasoning step follows the **same grammatical form** (e.g. 'X is true because... Y is true because...').
Chiasmus	Answer the question using a **mirrored reasoning structure**: first introduce reasoning elements **A then B**, then resolve them in reverse order (**B then A**) to reach your conclusion.

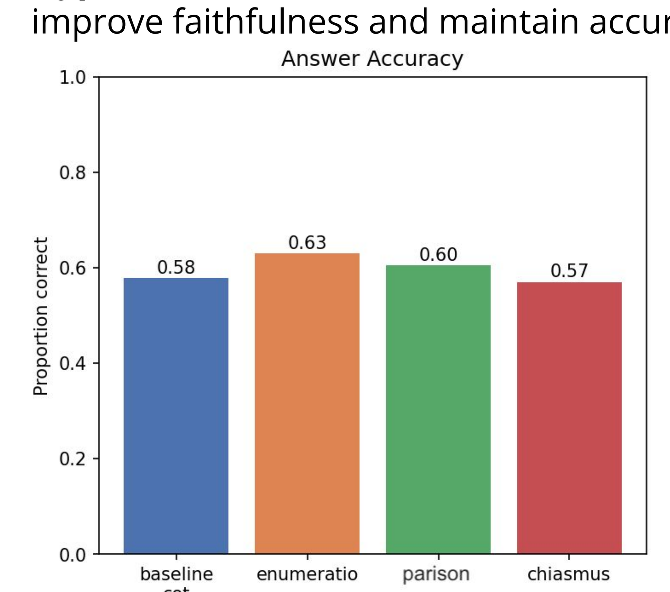
Metrics:
- Answer accuracy (yes/no vs. gold label)
- Faithfulness (1–5: does the reasoning chain logically support the answer?) - Claude Haiku
- Blind structure classification (is the figure identifiable from the output) - Claude Haiku

Probes: hidden states extracted at all 32 layers (last token). Logistic regression trained on 80/20 split. Direction vectors extracted from probe weights for Experiment C steering.

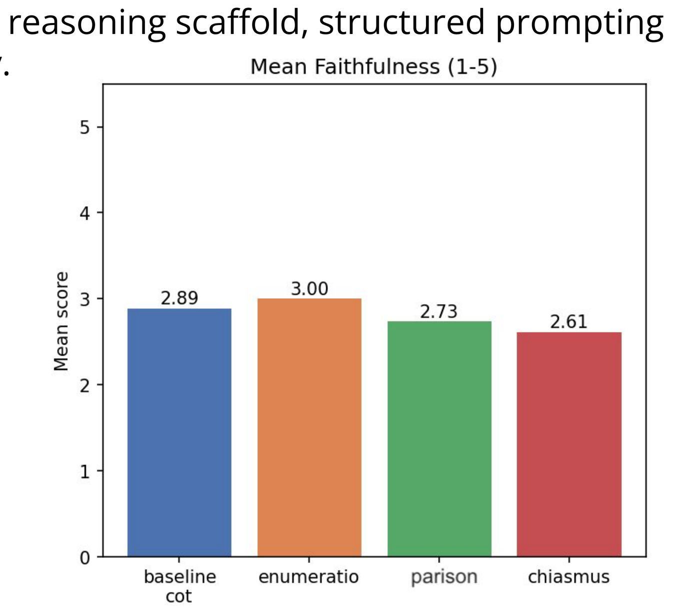
Github Repository Link (with experimental code)

Experiment A: Prompting

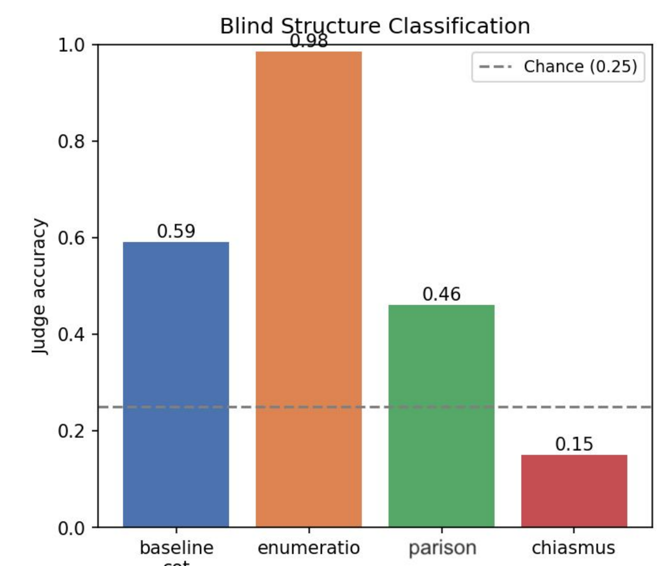
Hypothesis: If rhetorical structure acts as a reasoning scaffold, structured prompting should improve faithfulness and maintain accuracy.

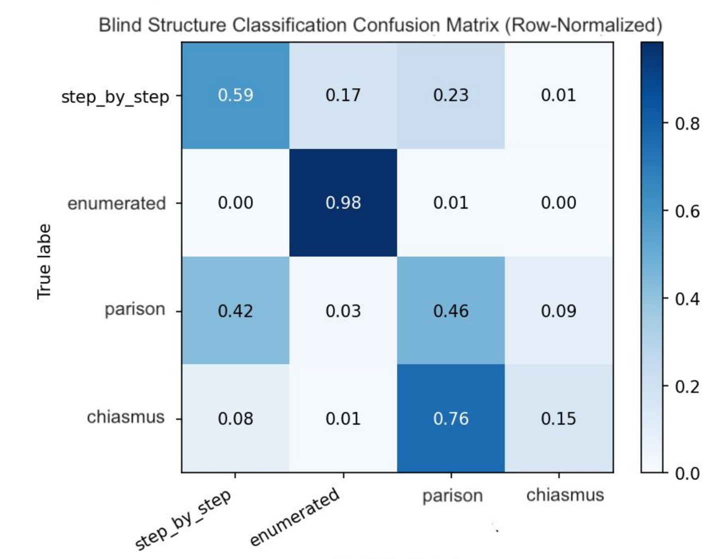


Enumeratio and Parison beat Baseline on Accuracy.



Enumeratio beat Baseline on Faithfulness. Chiasmus falls significantly short.



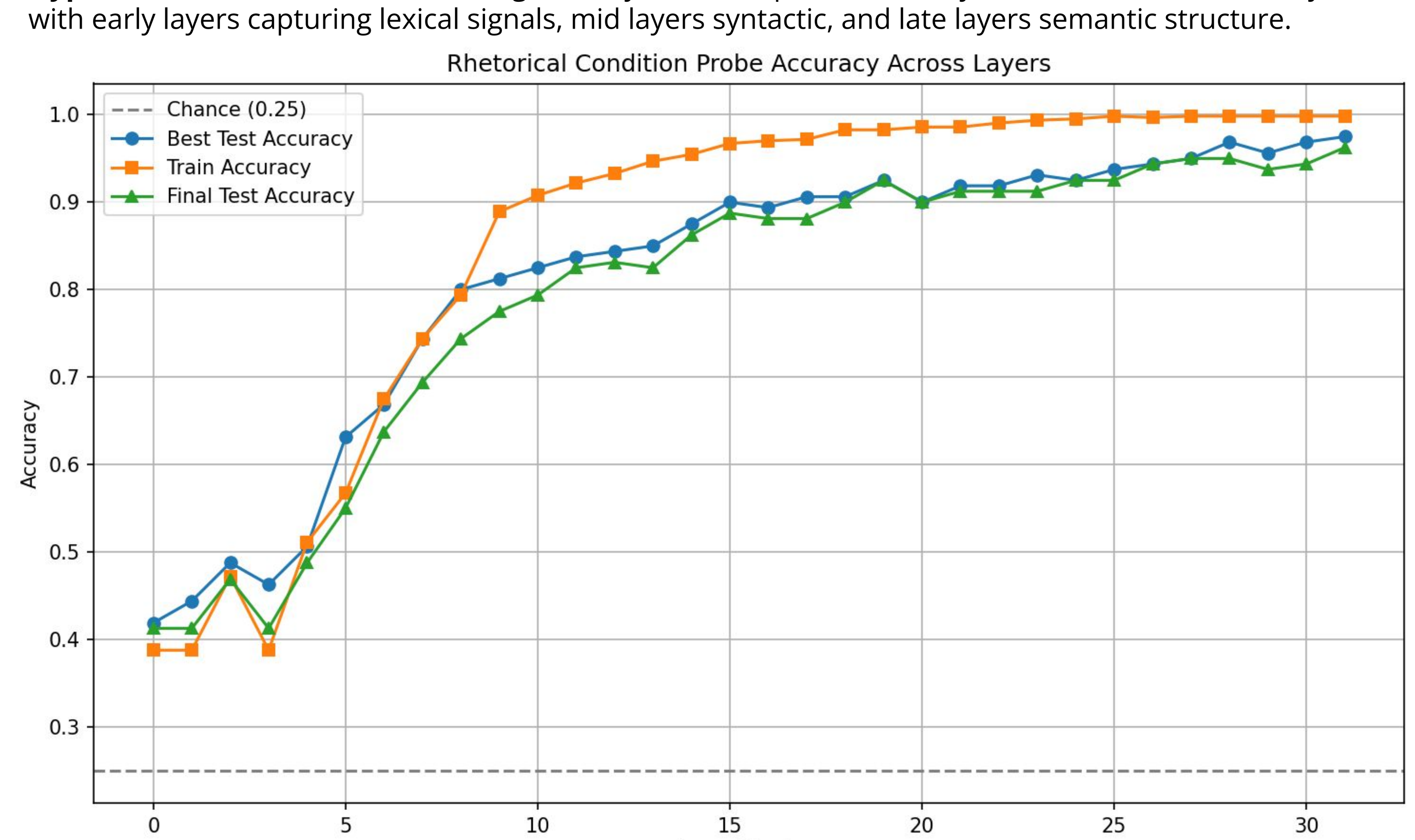


Enumeratio is highly identifiable due to its explicit lexical markers (1. 2. 3.). Chiasmus is most often confused for Parison (76% of the time): Harris [1] identifies that Antimetabole (subtype of chiasmus) co-occurs naturally with Parison in the AMP cluster, sharing communicative functions. If the model attempts chiasmus but fails to execute the inversion, the parallel structure that underlies both figures is what remains. Parison is often mistaken for baseline cot.

Experiment B: Probing

Using the reasoning traces generated in Experiment A, hidden states were extracted at each of Llama-2's 32 layers (last token, all layers). Linear logistic probes were trained to classify which rhetorical condition produced each trace.

Hypothesis: If rhetorical structure is genuinely encoded, probe accuracy should increase across layers, with early layers capturing lexical signals, mid layers syntactic, and late layers semantic structure.



Our rhetorical figures are linearly separable in mid-to-late layers. A simple logistic probe reaches ~97% test accuracy by layer 27, meaning the four conditions are not merely distinguishable but geometrically spread apart in activation space.

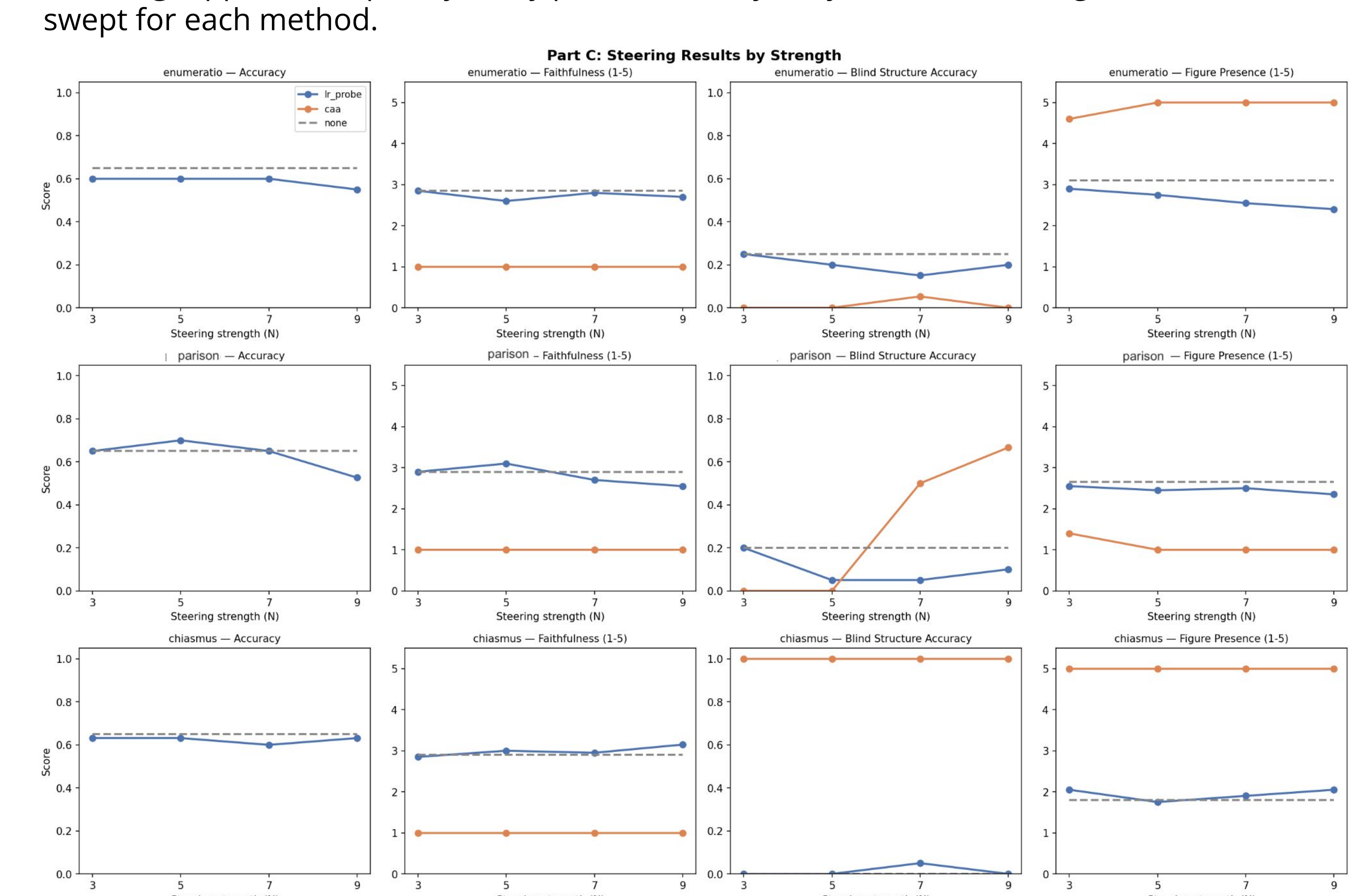
Chiasmus: Below-chance blind structure classification in Part A (0.15) but high probe accuracy in Part B. Since chiasmus outputs collapsed into parison, the probe may be tracking prompt-level features rather than true chiasmus geometry.

Controls: To test whether probe accuracy reflects rhetorical structure or just response length, traces were truncated to the length of the shortest condition before re-extracting activations and re-training probes. Length-matched probes plateau at ~75% .True accuracy lies between this 75% the 97% upper bound.

Experiment C: Steering

Using direction vectors extracted from Part B probes, hidden states were steered during generation to test whether pushing activations toward a rhetorical condition changes reasoning quality.

Steering: Applied to top-5 layers by probe accuracy (~layers 24-31). Strengths N in {3, 5, 7, 9} swept for each method.



- **Parison at N=5**: LR probe steering improves accuracy from 0.65 to 0.70 and faithfulness from 2.90 to 3.10 - the clearest evidence that activation steering can causally shift reasoning quality via rhetorical direction vectors.
- **Limits**: Enumeratio accuracy declines monotonically with steering strength and collapses at N≥5. Chiasmus is entirely unresponsive, consistent with its collapsed representation in Part B.

Conclusions

	Enumeratio	Parison	Chiasmus
Beats baseline on accuracy	Yes (small)	Yes (small)	No
Beats baseline on faithfulness	Yes	No	No
Reliably elicited in output	Yes	Partial	No
Linearly separable in activations	Yes (surface-driven)	Yes	Yes (despite output failure)
Causally steerable	Fragile (collapses at N>=5)	Yes, at N=5	No

Rhetorical structure can improve reasoning quality and is causally steerable, but steerability requires elicitability, and figures the model cannot produce cannot be steered into. These results are preliminary but consistent, and suggest that tying LLM reasoning closely to rhetorical structure is a promising direction for future work.

References and Acknowledgements

I would like to thank Professor Randy Harris for introducing me to the Rhetoricon project in 2023 by offering me a summer internship as a developer. Since then, he has been a continuous source of encouragement, kindness, and guidance - including generous feedback on my ideas and support throughout this project. I would also like to thank Adeshola Ogunsanya for her friendship, which has allowed me to stay connected to the group since 2023, and Zoya Randhawa for her guidance with the poster.

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